



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD 6.GR.2

Volume of Rectangular Prisms

Lesson 10-2 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period



TODAY'S OBJECTIVES

Content Objective: I can find the volume of a rectangular prism, including ones with fractional edge lengths, using $\text{base area} \times \text{height}$.

Language Objective: I can explain my work using the words volume, rectangular prism, dimensions, and base area.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Volume

Rectangular prism

Cubic units

Dimensions

Net

Base area



Tap any word to see what it means and a picture.

1 The amount of space inside a three-dimensional solid is its .

2 A solid with six rectangular faces is a .

3 The unit used to measure volume is .

4 The measurements of length, width, and height of a solid are its .

5 A flat pattern that folds into a solid is a .

6 The area of the bottom face of a prism, multiplied by height to find volume, is the .

Reflect — Exit Ticket

A rectangular prism has $l = 7$ ft, $w = 2$ ft, $h = 3$ ft. What is the volume?

- A. 42 ft^3
- B. 24 ft^3
- C. 42 ft^2
- D. 12 ft^3

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Dimensions
5. **Notes 5:** Net
6. **Notes 6:** Base area
7. **Watch for this mistake:** *A common mistake in Volume of Rectangular Prisms is skipping the key idea: "A fractional edge does not change the rule: still multiply length $\times$ width $\times$ height." — always check your work against this rule before you submit.*
8. **Try It 1:** A. Box A, because 50 cubic in is more than 48 cubic in — Box A: $5 \times 5 \times 2 = 50$ cubic in. Box B: $4 \times 4 \times 3 = 48$ cubic in. $50 > 48$, so Box A holds more.
9. **Try It 2:** A. 18 cubic in — $V = 4 \times 3 \times 1.5 = 12 \times 1.5 = 18$ cubic in. *A fractional edge does not change the steps.*
10. **Exit Ticket:** A. 42 ft³ — $V = 7 \times 2 \times 3 = 42$ cubic feet. *Remember: volume uses cubic units (ft³).*